Cryptocurrency Fundamentals

Module 1 Study Guide

Introduction to Blockchain Technology

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: What a Blockchain Actually Is

In this lesson: Centralized vs Distributed Ledgers; How Nodes Verify Transactions; Worked TT$

Key points

- Distributed ledgers replace central authority databases
- Nodes verify balances before committing transactions
- Majority consensus prevents fraud and bad records

Examples from the lesson

- Because this is our very first lesson of Module 1, Introduction to Blockchain Technology, we are starting with the foundation of everything: Lesson 1, What a Blockchain Actually Is.
- That single notebook works well for one person, but if it gets lost, damaged, or altered, nobody else has a record to prove what really happened.
- A centralized system, like a traditional bank account in Port of Spain, operates similarly because one central authority holds the master record.
- Instead of one bank keeping the master book, every participating computer—called a node—holds an identical, updated copy of the exact same record.
- Imagine three friends—Kevon, Priya, and Dev—running a shared ledger network to track their transactions in Trinidad and Tobago dollars.
- Kevon starts his account balance with TT$1,000, Priya starts with TT$500, and Dev starts with TT$250.
- This initial distribution forms our starting ledger state across all three nodes.
- Now, Kevon decides to buy a lime pass for an upcoming Carnival fete from Priya for TT$300.
- Kevon broadcasts this transaction to the entire network: Kevon sends TT$300 to Priya.
- Every node receives this broadcast, checks Kevon’s current balance, and verifies that TT$1,000 is more than enough to cover the TT$300 transfer.
- After this transaction is verified and settled, Kevon’s ending balance decreases to TT$700.
- Priya’s balance increases from her original TT$500 by adding the TT$300 payment, bringing her new total to TT$800.
- Dev did not participate in the trade, so his balance remains unchanged at TT$250.
- The total value recorded across the entire system remains exactly TT$1,750, proving that

every dollar is fully accounted for without any central authority managing the calculation.

- If Kevon attempts to alter his local notebook to claim he still has TT$1,000, his balance immediately conflicts with the identical records stored by Priya and Dev.
- If one computer shuts down or loses internet connectivity during a thunderstorm in San Fernando, the remaining nodes keep operating without missing a beat.
- Transactions are verified by the network checking existing balances, such as updating Kevon's account to TT$700 and Priya's to TT$800, before broadcasting identical updates to every node.
- In our next lesson, Lesson 2: Blocks, Chains, and Consensus, we will look under the hood to see how transactions are bundled together into permanent blocks and linked chronologically.

Lesson 2: Blocks, Chains, and Consensus

In this lesson: Grouping Transactions; Linking Blocks; Network Consensus; Practical Security.

Key points

- Grouping transactions into verified blocks
- Linking blocks chronologically with digital signatures
- Reaching network consensus to lock the ledger

Examples from the lesson

- I’m glad you’re back with us for "Cryptocurrency Fundamentals," Module 1, "Introduction to Blockchain Technology."
- Today, we’re diving into Lesson 2: "Blocks, Chains, and Consensus."
- Suppose three friends are out at a Sunday fete in Chaguaramas settling up their lime expenses.
- First, Keshore sends TT$150 to Maya for his share of the entry tickets.
- Next, Maya transfers TT$80 to Devon for drinks at the bar.
- Finally, Devon sends TT$50 back to Keshore for the ride home.
- These three individual transactions are bundled together into a single block.
- In our fete pool, Keshore spent TT$150 and received TT$50, leaving him with a net outgoing total of TT$100.
- Maya received TT$150 and spent TT$80, giving her a net incoming balance of TT$70.
- Devon received TT$80 and spent TT$50, leaving him with a net incoming balance of TT$30.
- Notice how the total outgoing TT$100 matches the incoming TT$70 plus TT$30 perfectly.
- Block 2 holds the signature of Block 1, and Block 3 holds the signature of Block 2.
- Instead of trusting one central bank or credit union, hundreds of computer nodes across the globe check every transaction against identical copies of the ledger.
- Second, each block is verified and linked to the previous one using unique digital signatures.

Lesson 3: Why Blockchain Matters Beyond Crypto

In this lesson: Permanent Land Records; Efficient Business Payments; Worked Cost Comparison

Key points

- Permanent tamper-proof record-keeping for land deeds
- Drastically slashes international payment fees for local businesses
- Guarantees authentic tracking for supply chains from harvest to market

Examples from the lesson

- I am your instructor, and welcome to Cryptocurrency Fundamentals, Module 1, Introduction to Blockchain Technology.
- In Lesson 3, we are looking at Why Blockchain Matters Beyond Crypto.
- The cocoa processor transfers a purchase payment of TT$ 50,000 to the overseas vendor.
- Under standard wire methods, the local bank charges an international processing fee of TT$ 1,500, while intermediary routing costs add TT$ 1,000.
- That brings total administrative overhead to TT$ 2,500.
- When utilizing a blockchain-based payment network, the international processing fee drops to TT$ 200, and intermediary routing costs are reduced to TT$ 50.
- The total administrative overhead for the blockchain transfer comes out to TT$ 250.
- Subtracting the TT$ 250 blockchain cost from the TT$ 2,500 traditional cost leaves the business with a net savings of TT$ 2,250 on a single transaction.
- That extra TT$ 2,250 stays right in the business owner's Trinidad and Tobago bank account to reinvest in operations.

Recap checklist — can you say each one?

Lesson 1: What a Blockchain Actually Is

- ☐ Distributed ledgers replace central authority databases
- ☐ Nodes verify balances before committing transactions
- ☐ Majority consensus prevents fraud and bad records

Lesson 2: Blocks, Chains, and Consensus

- ☐ Grouping transactions into verified blocks
- ☐ Linking blocks chronologically with digital signatures
- ☐ Reaching network consensus to lock the ledger

Lesson 3: Why Blockchain Matters Beyond Crypto

- ☐ Permanent tamper-proof record-keeping for land deeds
- ☐ Drastically slashes international payment fees for local businesses
- ☐ Guarantees authentic tracking for supply chains from harvest to market